



TY Coding

Introduction



Learning Outcomes

- Understand what Python is
- Understand some key terms associated with Coding Python
- Create a simple python program
- Output text to the screen
- Setup and use an IDE (Thonny)



Where is Python Used

- Web Development
 - Spotify,
 - YouTube
- Artificial Intelligence
 - Hear – Detecting Gunshots and location
 - Smell – Detecting Illness by smelling breath
 - See – Machine Vision in Self Driving Cars
 - Write – The NY Times uses AI to assist in writing articles
- Scripting
 - A script is a small piece of code that can automate a computer Function



Why is python popular

- Readability: Python's syntax (the rules for writing code) is very clean and uses indentation to define structure, making it look a bit like plain English.
- Versatility: (Transition to the next section) It can be used for a huge variety of tasks.
- Large Standard Library: Python comes with a vast collection of pre-written code modules that perform common tasks
- Huge Community: Because it's so popular, there are tons of resources, tutorials, forums, and people willing to help if you get stuck.
- Open Source: Python is free to use and distribute, even for commercial purposes.



Examples

- `print("Hello, World!")`
- `print` is the command and tells python to place text on the screen
- `( )` What is to be printed goes in here
- `" "` Tells python that that it is text that is to be printed
- `print(10 + 5)`
- Python will carry out the sum and print the answer



Creating Code

- There are a number of ways but the most common is to use an IDE (Integrated Development Environment)
- This is an application that allows you to create code and then run it.
- Can be online or offline
- Online
 - Replit, Codepad.app, trinket.io
- Offline
 - Visual Studio Code, Thonny, IDLE
- Advantages and Disadvantages

Using IDLE



- Installs with python
- Allows you to test python code
- Allows you to create python files
- Has some limitations
- A dedicated IDE has more features



Using Thonny

- Thonny should be installed on every computer
- Can be found on the Desktop
- Can be downloaded from - <https://thonny.org/>
- Files are save to PC with extension .py



File Management

- Files are stored on the Computer
- Creating a Folder Structure to organise files is important
- Create a Folder in Share Drive
 - Found in My computer (\$User.Name)
 - This folder appears on every computer you log in to
 - Call Folder TY Coding
 - In this Create folder called Lesson 01
 - Save all files today to this Folder



Terms to Know (Printing)

- Character
 - Any individual letter, number or symbol e.g. a, 4, €
 - String
 - A group of 2 or more characters e.g. PYTHON is made up of Characters P Y
T H O N
 - Function
 - Another name for a command that python recognises e.g. print()

Using Thonny



- Thonny is a free IDE that runs on your computer
- Can be found at - <https://thonny.org/>
- Can be installed or run from a portable folder
- Comes with common Python libraries
- More can be installed as needed.



- Print a string to the screen
 - We use the print() command
 - We put what we want to print in brackets
 - String must have “ “
 - print(“Hello World”)

Note:
Functions in Python are case sensitive
print is the function to output to the screen
Print will give an error message



- Create a file system in your shared drive
 - Folder for Coding
 - Create a folder called Lesson 01
 - This will contain all the files for this lesson
- Open Thonny and create a new file
- Save the file to a folder for Coding
- Call the file Lesson 01 printing.py



- Print the following
 - I'm learning to code
I'm using python
- This needs 2 lines of code
 - `print("I'm learning to code")`
 - `print("I'm using Python")`
- Alternative is to use a special command for a new line `\n`
- `print("I'm learning to code \nI'm using Python")`



- ◆ Print the following Using the print command
- ◆ * There are Spaces on the Screen *
- ◆ You can use \t to add in a tab space which is the same as 3-5 spaces
- ◆ Use the following command
- ◆ `print("* \tThere are Spaces on the Screen\t*")`

Summary



- This lesson covered
 - Characters
 - Strings
 - Functions
 - Introducing Thonny
 - How to use the print function
 - Some special characters